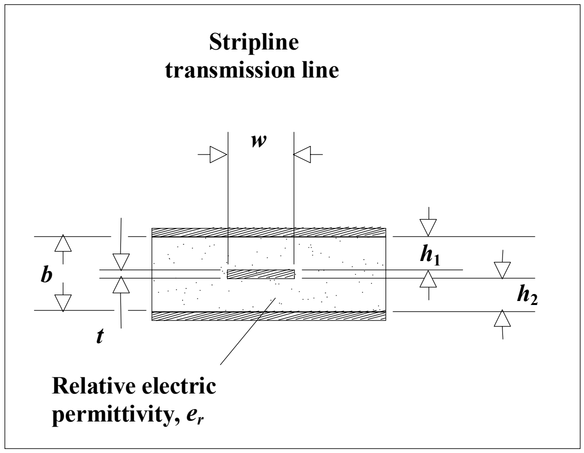
STRIPLINE TRANSMISSION LINES                    file:  sline.mcd

Formulas included in this spreadsheet:

| | |
|---|---|
| Stripline characteristic impedance | ZSTRIP() |
| Offset stripline characteristic impedance | ZOFFSET() |
| Stripline propagation delay | PSTRIP() |
| Stripline trace inductance | LSTRIP() |
| Offset stripline inductance | LOSTRIP() |
| Stripline trace capacitance | CSTRIP() |
| Offset stripline capacitance | COSTRIP() |

Formulas are from Seymour Cohn, "Problems in Strip Transmission Lines," MTT-3, No. 2, March 1955, pp. 199-126.

This material is summarized in Harlan Howe, Stripline Circuit Design,  Artech House, Norwood, MA, 1974.




sline

Variables used:

| | |
|---|---|
| h1 | Trace height above lower ground plane (in.) |
| h2 | Trace headroom below upper ground plane (in.) |
| b | Separation between ground planes, b = h1 + h2 + t (in.) |
| w | Trace width (in.) |
| t | Trace thickness (in.) |
| er | Trace thickness (in.) |
| x | Trace length (in.) |

Stripline characteristic impedance (Ω:)

Accuracy of better than 1.3% is
obtained under the following conditions:

```
t/b  <  0.25
t/w  <  0.11
er   unrestricted
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NOTE: formula ZSTR_K1()
corrected per instructions
from Robert Canright of
Richardson, TX.  Thanks, Robert.

For skinny traces  (w/b < 0.35)

$$ZSTR\_K1(w,t) := \left(\frac{w}{2}\right)\cdot\left[1+\frac{t}{\pi\cdot w}\cdot\left(1+\ln\left(\frac{4\cdot\pi\cdot w}{t}\right)\right)+0.255\cdot\left(\frac{t}{w}\right)^2\right]$$

$$ZSTR\_skny(b,w,t,er) := \frac{60}{\sqrt{er}}\cdot\ln\left(\frac{4\cdot b}{\pi\cdot ZSTR\_K1(w,t)}\right)$$

For wide traces  (w/b > 0.35)

$$ZSTR\_K2(b,t) := \left[\frac{2}{1-\frac{t}{b}}\cdot\ln\left(\frac{1}{1-\frac{t}{b}}+1\right)-\left(\frac{1}{1-\frac{t}{b}}-1\right)\cdot\ln\left[\frac{1}{\left(1-\frac{t}{b}\right)^2}-1\right]\right]$$

$$ZSTR\_wide(b,w,t,er) := \frac{94.15}{\frac{\frac{w}{b}}{1-\frac{t}{b}}+\frac{ZSTR\_K2(b,t)}{\pi}}\cdot\frac{1}{\sqrt{er}}$$

Composite formula picks skinny or
wide model depending on w/b ratio:

$$ZSTRIP(b,w,t,er) := \mathrm{if}(w > .35\cdot b, ZSTR\_wide(b,w,t,er), ZSTR\_skny(b,w,t,er))$$

Rarely are the two parameters h1 and h2 equal in practice. The more common case is an assymetric stripline having the conducting trace offset to one side.

Offset, or asymmetric, stripline characteristic impedance (Ω) (no accuracy guaranteed):

$$\mathrm{ZOFFSET}(h1,h2,w,t,er) := \frac{2\cdot \mathrm{ZSTRIP}(2\cdot h1+t,w,t,er)\cdot \mathrm{ZSTRIP}(2\cdot h2+t,w,t,er)}{\mathrm{ZSTRIP}(2\cdot h1+t,w,t,er)+\mathrm{ZSTRIP}(2\cdot h2+t,w,t,er)}$$

Propagation delay of stripline (s/in.):

$$\mathrm{PSTRIP}(er) := 84.72\cdot 10^{-12}\cdot\sqrt{er}$$

(same formula for centered or offset stripline)

Inductance of stripline (H):

$$\mathrm{LSTRIP}(b,w,t,x) := \mathrm{PSTRIP}(1.)\cdot \mathrm{ZSTRIP}(b,w,t,1.)\cdot x$$

In the equation above, we can assume a relative permittivity of 1.; it doesn't affect the answer.

Inductance of offset stripline (H):

$$\mathrm{LOSTRIP}(h1,h2,w,t,x) := \mathrm{PSTRIP}(1.)\cdot \mathrm{ZOFFSET}(h1,h2,w,t,1.)\cdot x$$

Capacitance of stripline (F):

$$\mathrm{CSTRIP}(b,w,t,er,x) := \frac{\mathrm{PSTRIP}(er)}{\mathrm{ZSTRIP}(b,w,t,er)}\cdot x$$

In the equations above and below, we must use the relative permittivity.

Capacitance of offset stripline (F):

$$\mathrm{COSTRIP}(h1,h2,w,t,er,x) := \frac{\mathrm{PSTRIP}(er)}{\mathrm{ZOFFSET}(h1,h2,w,t,er)}\cdot x$$

## Example stripline calculations

Ground plane separation (in.) $B := .020$

Width of trace (in.) $W := .006$

Thickness of trace (in.) $T := .00137$ (1-oz copper plating weight)

Length of wire (in.) $X := 11.000$

Relative electric permeability (affects capacitance, but not inductance) $er := 4.5$

Impedance (Ω):

$$ZSTRIP(B, W, T, er) = 51.4371$$

Total inductance (H):

$$LSTRIP(B, W, T, X) = 1.0169 \times 10^{-7}$$

Same result in nH:

$$LSTRIP(B, W, T, X) \cdot 10^{9} = 101.686$$

Inductance per in. (H):

$$LSTRIP(B, W, T, 1) = 9.2442 \times 10^{-9}$$

Total capacitance (F):

$$CSTRIP(B, W, T, er, X) = 3.8433 \times 10^{-11}$$

Same result in pF:

$$CSTRIP(B, W, T, er, X) \cdot 10^{12} = 38.4334$$

Capacitance per in. (F):

$$CSTRIP(B, W, T, er, 1) = 3.4939 \times 10^{-12}$$

Tolerance effects

$$\mathrm{ZOFF\_TOL}(h1,dh1,h2,dh2,w,dw,t,er,der) := \begin{pmatrix} \mathrm{ZOFFSET}(h1+dh1,h2+dh2,w-dw,t,er-der) \\ \mathrm{ZOFFSET}(h1,h2,w,t,er) \\ \mathrm{ZOFFSET}(h1-dh1,h2-dh2,w+dw,t,er+der) \end{pmatrix}$$

$$\alpha := \mathrm{ZOFF\_TOL}(.007,.002,.032,.002,.008,.002,.0015,4.5,.1)$$

$$\alpha = \begin{pmatrix} 64.0566 \\ 51.7263 \\ 39.228 \end{pmatrix}$$

$$\mathrm{REFL}(x,z) := \begin{pmatrix} \dfrac{z-x_0}{z+x_0} \\ \dfrac{z-x_1}{z+x_1} \\ \dfrac{z-x_2}{z+x_2} \end{pmatrix}$$

$$\mathrm{REFL}(\alpha,50) = \begin{pmatrix} -0.1232 \\ -0.017 \\ 0.1207 \end{pmatrix}$$